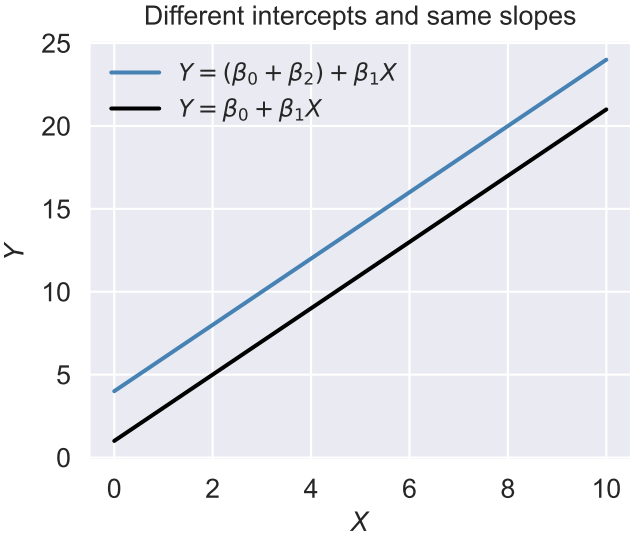
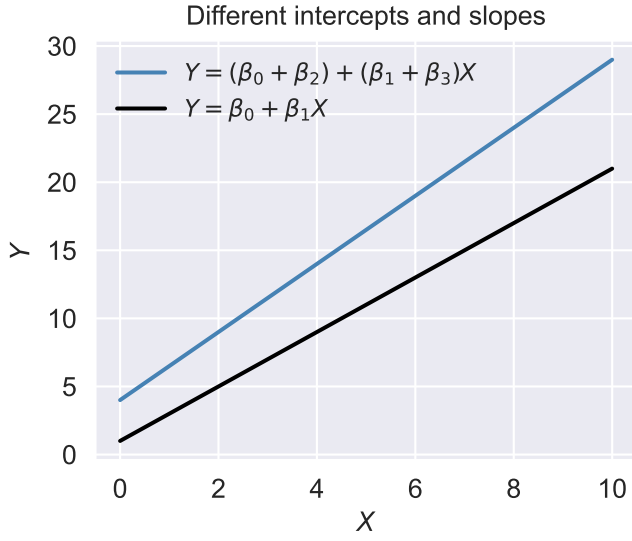
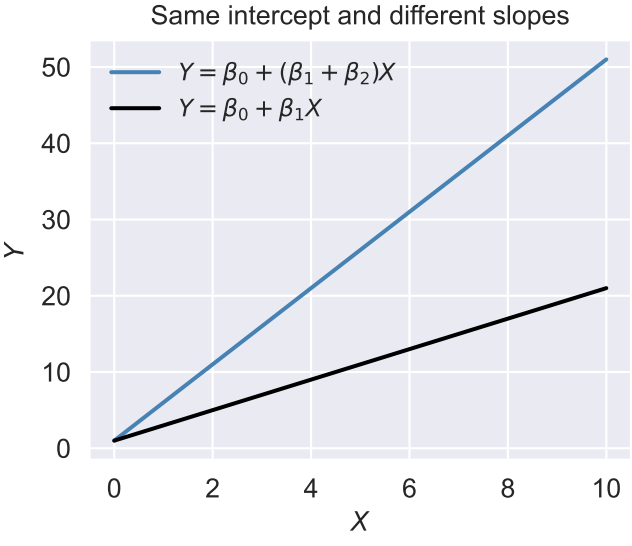

Different intercepts and same slopes
$Y = (\beta_0 + \beta_2) + \beta_1 X$
$Y = \beta_0 + \beta_1 X$
Y
X
0
5
10
15
20
25
0
2
4
6
8
10


Different intercepts and slopes
$Y = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) X$
$Y = \beta_0 + \beta_1 X$
Y
X
0
5
10
15
20
25
30
0
2
4
6
8
10


Same intercept and different slopes
$Y = \beta_0 + (\beta_1 + \beta_2) X$
$Y = \beta_0 + \beta_1 X$
Y
X
0
10
20
30
40
50
0
2
4
6
8
10